

Project Initialization and Planning Phase

Date	10 July 2026
Team ID	
Project Title	Raising Waters
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report

The proposal report aims to transform loan approval using machine learning, boosting efficiency and accuracy. It tackles system inefficiencies, promising better operations, reduced risks, and happier customers. Key features include a machine learning-based credit model and real-time decision-making.

Project Overview	
Objective	The primary objective is to revolutionize flood management and early warning capabilities by implementing advanced machine learning techniques, ensuring faster, highly accurate regional flood risk tier assessments.
Scope	The project comprehensively assesses incoming meteorological parameters, incorporating automated real-time data ingestion pipelines and specialized machine learning evaluation models for a robust predictive early-warning architecture.
Problem Statement	
Description	Addressing delayed alerts, fragmented sensor inputs, and manual data computation inefficiencies in current disaster mitigation systems, which severely compromises emergency response windows and community safety.
Impact	Solving these critical issues will result in significantly improved proactive evacuation protocols, minimized infrastructure damage, reduced public risk, and a drastic increase in automated early warning execution success rates.
Proposed Solution	
Approach	Employing machine learning classification algorithms (such as XGBoost/KNN) to analyze historical and live atmospheric data metrics, creating a dynamic, instantly adaptable risk tier evaluation pipeline.
Key Features	-- Automatic data ingestion linking external live Weather APIs straight to ML processing elements. - Real-time predictive evaluation that dynamically outputs color-coded risk levels.

	- Centralized open architecture via a responsive Flask Dashboard that registers continuous historical prediction logs for future verification.
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Resource Requirements

Resource Type	Description	Specification/Allocation
Hardware		
Computing Resources	CPU/ cores	Standard Quad-Core Processor (Intel i5/ AMD Ryzen 5 or equivalent)
Memory	RAM specifications	8 GB /16 GB RAM
Storage	Disk space for data, models, and logs	256 GB / 512 GB SSD (Local Storage for .pkl models & log files)
Software		
Frameworks	Web/Application frameworks	Flask
Libraries	Additional libraries	scikit-learn, pandas, numpy, matplotlib, seaborn
Development Environment	IDE	Jupyter Notebook, pycharm
Data		
Data	Source, size, format	Historical Meteorological Flooding Datasets, Regional Monsoon Records, .csv format